

Ugochukwu Okonkwo

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PROFESSIONAL SUMMARY

Biomedical Engineer with hands-on experience in medical device design, CAD modeling, prototyping, usability testing, and regulated product development. Experienced in supporting orthopedic implant and surgical instrument development, DV&V activities, engineering documentation, and cross-functional collaboration within MedTech environments. Combines technical engineering expertise with business and project management knowledge to support the development of innovative, patient-centered medical devices.

RELEVANT WORK EXPERIENCE

Globus Medical, Audubon, PA

Project Engineer Co-op, Jan 2024 - Aug 2024

- Applied Creo Parametric and engineering drawing standards to support development of orthopedic implants and surgical instruments in a regulated MedTech environment
- Utilized rapid prototyping, DFM principles, and cross-functional collaboration to support design iterations, manufacturability improvements, and functional evaluations
- Supported DV&V activities, engineering change processes, and technical documentation aligned with FDA and MDR quality system requirements
- Developed Excel macro-based inspection and traceability tools to improve quality documentation efficiency and workflow standardization
- Coordinated with manufacturing, operations, prototype shops, and suppliers to support prototype builds, testing, and production readiness
- Applied product lifecycle management concepts including DCO processing, UDI tracking, and DHF/DMR documentation support

EDUCATION

Florida Institute of Technology, Melbourne, FL

Ph.D. in Biomedical Engineering, Expected May 2030

Widener University, Chester, PA

Master of Science in Engineering in Biomedical Engineering (Thesis), May 2026

Thesis: Ergonomic Redesign of an Assistive Button Hook for Individuals with Dexterity Impairments

Master of Business Administration in Management, May 2026

Bachelor of Science in Biomedical Engineering, May 2025

Minors: Management, Mathematics, Mechanical & Robotics Engineering

CONFERENCE PRESENTATIONS

Biomedical Engineering Society (BMES) Annual Meeting, Orlando, FL | October 2026

Human-Centered Redesign of a Button Hook for Individuals with Dexterity Impairments: Development and Preliminary Usability Evaluation
Poster Presentation.

RESEARCH EXPERIENCE

Master's Thesis Research

Widener University | Aug 2025 – Jul 2026

- Designed and developed an ergonomic assistive medical device from concept through prototype evaluation, applying human factors engineering and user-centered design principles to improve usability for individuals with hand dexterity impairments.
- Generated and evaluated multiple design concepts using CAD, rapid prototyping, and iterative engineering reviews to optimize ergonomics, manufacturability, and functional performance.
- Developed custom manufacturing fixtures, wire-forming molds, and assembly processes using FDM 3D printing, CO₂ laser cutting, and precision fabrication techniques to improve prototype consistency.
- Conducted design verification through expert evaluations and structured usability testing, utilizing QUEST 2.0-based surveys and user feedback to assess comfort, ease of use, grip stability, and mechanism performance.
- Collaborated with interdisciplinary clinical stakeholders, including Physical Therapy and Occupational Therapy professionals, to translate user needs into measurable engineering design requirements.
- Analyzed quantitative and qualitative usability data to validate design decisions, guide iterative improvements, and support recommendations for future assistive device development.

Low-Cost Versatile Electrospinning System

Undergraduate Senior Research Project, Widener University | Aug 2024 – April 2025

- Applied CAD modeling, prototype fabrication, and mechanical assembly techniques to develop a low-cost electrospinning platform for nanofiber production
- Utilized bandsaws, table saws, and fabrication tools to machine acrylic panels and aluminum structural components for custom system integration
- Performed aluminum tapping, fastening, and enclosure assembly operations to support structural stability and modular component integration
- Incorporated iterative prototyping, 3D printing, and system integration methods to refine syringe pump and rotating collector assemblies
- Conducted experimental parameter optimization involving flow rate, collector speed, and polymer delivery to improve fiber alignment and consistency
- Implemented safety-focused engineering controls including insulated acrylic housing, grounding, and controlled high-voltage operation

ADDITIONAL WORK EXPERIENCE

Widener University Office of Student Success, Chester, PA

Graduate Student Employee, Aug 2025 - Jun 2026

- Coordinated tutor-student matching and supported front desk operations, improving service efficiency, scheduling, and student support delivery.
- Trained and onboarded new tutors, reinforcing academic support protocols and effective communication strategies.
- Analyzed tutoring and engagement data using Microsoft Excel (pivot tables, queries) and developed promotional materials to increase program effectiveness and student participation.

Peer & Math Tutor, Sep 2024 - May 2025

- Delivered one-on-one and group tutoring across courses (MGT-210, MATH-242, ENGR-213), adapting teaching strategies to diverse learning styles.
- Simplified complex mathematical and technical concepts while guiding students through problem-solving, projects, and exam preparation.
- Fostered student confidence, independent learning, and problem-solving skills.

Widener University Office of Admissions, Chester, PA

Crew Leader, Aug 2023 - Dec 2023

- Led orientation programs and delivered presentations to groups of 20+ students, supporting their transition into university life.
- Collaborated with admissions and student affairs teams during university events and outreach initiatives

CERTIFICATIONS

Certified SOLIDWORKS Professional (CSWP)

Certified SOLIDWORKS Sheet Metal Professional (CSWPA-SM)

PTC University Creo User Modeling Specialist Certification

PTC University Creo User Documentation Specialist

ISO 14971: Medical Device Risk Management

Medical Device Process Validation (ISO 13485, IQ/OQ/PQ)

Lean Six Sigma Green Belt

TECHNICAL SKILLS

- CAD & Design: Creo Parametric, Autodesk Inventor, SolidWorks, 3D Printing, Iterative Design, Prototyping
- Modeling & Simulation: ANSYS Fluent, Thermal & Bioheat Modeling, Mesh Independence Studies
- Data & Analysis: MATLAB, Excel (Regression, Pivot Tables, Data Analysis ToolPak), SQL, Tableau
- Engineering & Medical Device: Design Verification & Validation (V&V), Human-Centered Design, Ergonomics, Usability Testing, Quality Systems (QMS), CO₂ Laser Cutting, Bandsaw & Table Saw Operation
- Quality & Regulatory: Lean Six Sigma, DMAIC, FMEA, Process Improvement, Design Controls
- Laboratory Techniques: Bradford Assay, Protein Extraction (RIPA Buffer)

PROFESSIONAL SKILLS

Research communication • Technical writing • Project management • Mentorship • Leadership • Collaboration & Teamwork • Problem-solving & Critical thinking